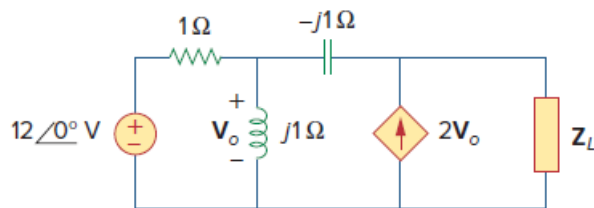


Problem

In the circuit of Fig. 11.46, find the value of Z_L that will absorb the maximum power and the value of the maximum power.

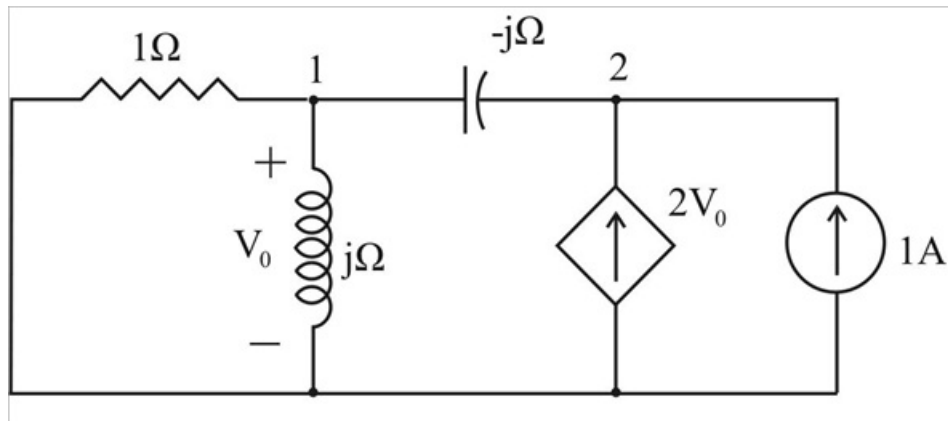
Figure 11.46:



Step-by-step solution

Step 1 of 8

To find Z_{TH} , insert a 1 – A current source at the load terminal as shown in fig (a).



Step 2 of 8

Writing nodal equations at the top nodes,

At node 1,

$$\frac{V_o}{1} + \frac{V_o}{j} = \frac{V_2 - V_o}{-j}$$

$$V_o = jV_2 \quad \text{.....(1)}$$

At node 2,

$$1 + 2V_o = \frac{V_2 - V_o}{-j}$$

$$1 = jV_2 - (2+j)V_o \quad \text{.....(2)}$$

Substituting (1) into (2),

$$1 = jV_2 - (2+j)(j)V_2$$

$$1 = (1-j)V_2$$

$$V_2 = \frac{1}{1-j}$$

Step 3 of 8

$$Z_{TH} = \frac{V_2}{1}$$

$$Z_{TH} = \frac{1+j}{2}$$

$$Z_{TH} = 0.5 + j0.5$$

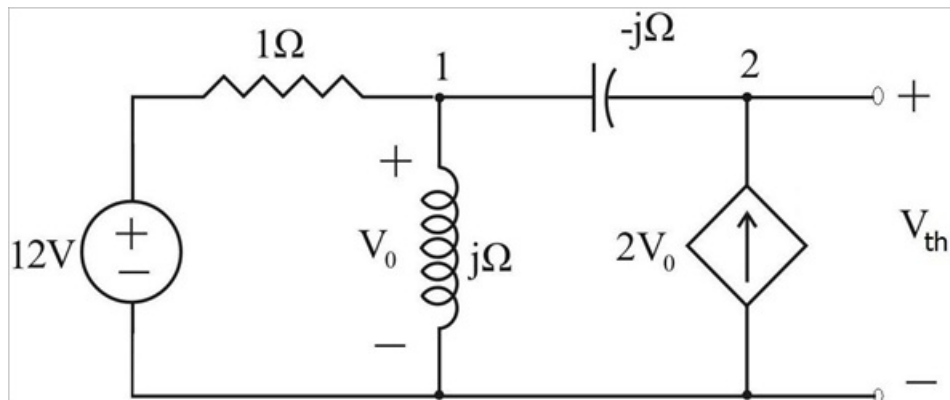
For maximum power transfer

$$Z_L = Z_{TH}^*$$

$$Z_L = 0.5 - j0.5 \Omega$$

Step 4 of 8

Now the Thevenin's equivalent voltage across the open circuit terminals



Step 5 of 8

Writing nodal equations at the top nodes,

At node 1,

$$\frac{V_0 - 12}{1} + \frac{V_0}{j} = \frac{V_2 - V_0}{-j}$$

$$V_0 - jV_2 = 12 \quad \dots\dots\dots(1)$$

At node 2,

$$2V_0 = \frac{V_2 - V_0}{-j}$$

$$(1 - 2j)V_0 - V_2 = 0 \quad \dots\dots\dots(2)$$

Step 6 of 8

Substituting (1) into (2),

$$(1 - 2j)(12 + jV_2) - V_2 = 0$$

$$12 + jV_2 - j24 + 2V_2 - V_2 = 0$$

$$12 - j24 + V_2(1 + j1) = 0$$

$$V_2 = \frac{-12 + j24}{1 + j1}$$

$$V_2 = \frac{(-12 + j24)(1 - j1)}{(1 + j1)(1 - j1)}$$

Step 7 of 8

$$V_2 = \frac{12 + j36}{2}$$

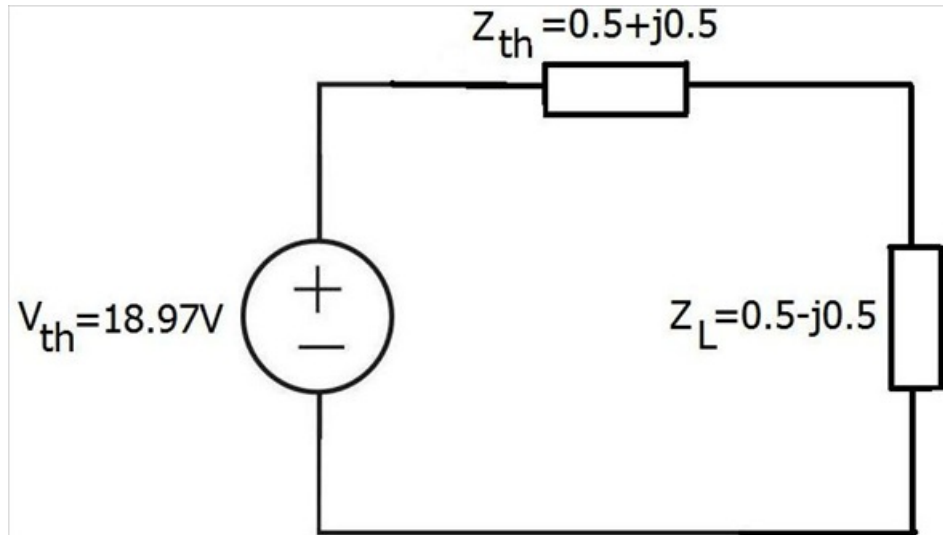
$$V_2 = 6 + j18 \text{ V}$$

$$V_2 = 18.97 \angle 71.57^\circ \text{ V}$$

Since $V_2 = V_{th}$

$$V_{th} = 18.97 \angle 71.57^\circ \text{ V}$$

Therefore the Thevenin's equivalent circuit is

**Step 8** of 8

Now the maximum power transferred is given by,

$$P_{\max} = \frac{V_{th}^2}{8R_L}$$

$$P_{\max} = \frac{(18.97)^2}{8(0.5)}$$

$$P_{\max} = 89.96 \text{ W}$$

$$\boxed{P_{\max} = 90 \text{ W}}$$